

AKUL TYAGI

Delhi, India | akultyagi.cs@gmail.com | (+91) 7503898231
[GitHub](#) | [Portfolio](#) | [LinkedIn](#) | [Codeforces](#): **1540+** (Specialist-top 7%)

TECHNICAL SKILLS

- Programming Languages: **Java, Kotlin, Python, JavaScript (ES6+), TypeScript, SQL.**
- Frontend: **React.js, Next.js (SSR/ISR), Three.js, TailwindCSS, HTML5, CSS3, Zustand, Framer Motion.**
- Mobile Development: **Android SDK, Jetpack Compose, MVVM, Coroutines, Room, Gradle, Retrofit.**
- Backend & DB: **Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, Redis, Microservices.**
- AI/ML: **PyTorch, Hugging Face Transformers, LLM Fine-tuning(LoRA/QLoRA), RAG Pipeline.**
- DevOps & Tools: **Git, Docker, Kubernetes, AWS(EC2,S3), Firebase, CI/CD, Github, Colab, Android Studio**
- Core CS: **Data Structures & Algorithm, Object-Oriented Design (LLD), System Design, DBMS, OS.**

EDUCATION

- B.TECH CSE | Galgotias College of Engineering and Technology (2023-2027)
- X (CBSE) | XII (CBSE) | JEE(Mains) 93.2% | 83.6% | 89.1%ile | 2021 | 2023

WORK EXPERIENCE

Naivety (Incubated Startup—GCET Accelerator) | Founder & Lead Android Engineer | Nov 2025 - Present

- Architected & Published an offline-first Android application using MVVM, Jetpack Compose, and Room Database, serving 700+ active users across 50+ countries with 20k+ 99.8% crash-free sessions by implementing Kotlin Coroutines, Retrofit and Dagger-Hilt (verified by Firebase Analytics & Android Vitals), on the [Play Store](#)
- Engineered a custom high-performance PDF rendering engine by forking, rewriting and rebuilding native C++ libraries (Github: [AndroidPdfViewer](#) & [PdfiumAndroid](#)) to enforce 16KB page-size alignment, reducing application memory footprint with 30% and ensured forward compatibility with Android 15 kernel mandates.
- Selected as 1 of 10 ventures from 300+ applicants for the GCET Startup Accelerator – Cohort 1 and to work under successful startup founders, as mentors, 1-on-1 for 6 months to scale Naivety ([Github](#)).

PROJECTS

Unagi - AI-Powered LinkedIn Content Generation Platform | [Github](#) | [Live Demo](#)
(Python, PyTorch, Falcon3-7B-Instruct, QLoRA, FastAPI, Next.js, Docker, Redis)

- Fine-tuned a Falcon3-7B-Instruct LLM on 1.2k+ high- engagement posts using QLoRA and 4-bit quantization; optimized inference on T4 GPUs to generate human-like content with 35% lower latency
- Built a scalable FastAPI backend with Celery and Redis for asynchronous generation queues; integrated a Next.js frontend with regex-based post-processing, driving 1,200+ impressions and reaching 1,800+ members.

CureAPT - Full-Stack Algorithmic Wellness & Data Engine | [Github](#) | [Live Demo](#)
(Next.js, TypeScript, MongoDB, Serverless, Node.js, Vercel)

- Architected a weighted-decision algorithm to normalise 30+ heterogeneous user inputs into a unified wellness index, utilising O(1) constant-time lookups for dynamic scoring, and a rule engine with reservoir sampling logic to generate personalised, tier-based feedback loops from 100+ actionable insights, ensuring high user retention.
- Optimised database throughput for serverless edge deployment by implementing global connection pooling for Mongoose, preventing connection exhaustion and ensuring fail-fast reliability during high-concurrency bursts.

ACHIEVEMENTS & CERTIFICATIONS

- **NPTel - Deep Learning (IIT Ropar): Elite + Silver Medal [Certificate](#)** (75%) in 12-week course covering neural networks, optimisation (SGD/Adam), regularisation and autoencoders, PCA/SVD, CNNs, RNNs/LSTMs.
- **Open Source Contribution:** One of the first globally to modernise AndroidPdfViewer & PdfiumAndroid libraries (Github: [AndroidPdfViewer](#) & [PdfiumAndroid](#)); Implemented 16KB page support & rebuilt 64-bit libs to comply with Google Play 2025 policies.
- **Codeforces Specialist (1540+ rating):** Ranked in top 7% globally; solved 350+ DSA problems across platforms.